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# M101 · Unit 3 · Quick reference

Matrices, transformations and simultaneous equations

v1.0 · 12 October 2026 · Recall / Understand / Consult

## Recall · Dimensions and columns

$$A=\begin{pmatrix}a & b\\ c & d\end{pmatrix},\quad A\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}ax+by\\ cx+dy\end{pmatrix}$$

m rows × n columns: n input entries, m output entries. Entry $a_{ij}$: row i, column j. Coordinate-vector images are the columns.

## Recall · Composition

$$(BA)\mathbf{v}=B(A\mathbf{v})$$

Rightmost acts first. Inner dimensions must match. AB and BA can differ. Identity I leaves every compatible vector unchanged.

## Consult · Two-by-two product

$$\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}e & f\\ g & h\end{pmatrix}=\begin{pmatrix}ae+bg & af+bh\\ ce+dg & cf+dh\end{pmatrix}$$

Each output entry combines one row of the left matrix with one column of the right matrix.

## Understand · Solution-preserving operations

$$R_i\leftrightarrow R_j,\quad R_i\leftarrow kR_i\ \left(k\neq 0\right)$$

$$R_i\leftarrow R_i+kR_j\quad \left(i\neq j\right)$$

Operate on whole equations, including right-hand sides. Keep the other row in a row replacement. Each operation is reversible.

## Understand · Elimination and solution sets

$$0=c\ (c\neq 0)\Rightarrow \text{no solution}$$

0 = 0 adds no condition. Keep the remaining equations. Back-substitute when each unknown is fixed; otherwise describe allowed free parameters. Verify against the originals.

## Understand · A complete parameter family

$$x+2y=6\iff (x,y)=(6-2t,t)$$

t ranges over all real numbers. Check both directions: every parameter gives a solution; every solution has such a parameter. Sample pairs alone do not prove completeness.

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## Recall · Determinant and inverse

$$\Delta = ad - bc, \quad A^{-1} = \frac{1}{\Delta}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Inverse formula requires Δ ≠ 0. Then every target has exactly one input. Check $AA^{-1} = A^{-1}A = I$. With Δ = 0, inspect the target: no solution or infinitely many.

## Understand · Area and orientation in the plane

$$\text{image area} = |\Delta| \text{ original area}$$

Same perpendicular, equal-scale coordinates. Positive Δ preserves orientation; negative reverses it; zero collapses area. Area scale does not specify length or angle changes.

## Understand · Residuals and their limits

$$\mathbf{r} = A\mathbf{x}_{\text{candidate}} - \mathbf{b}$$

Every entry zero: candidate satisfies these equations. This does not prove uniqueness. A failed candidate does not prove inconsistency. Small residual need not mean small input error.

## Consult · Transparent Python check

```
def residual2(m, v, target):
    first = m[0]*v[0] + m[1]*v[1]
    second = m[2]*v[0] + m[3]*v[1]
    return [first-target[0], second-target[1]]

r = residual2([1, 2, 0, 1], [1, 2], [5, 2])
ok = r[0] == 0 and r[1] == 0
print(r, ok)  # [0, 0] True
```

m = [a,b,c,d] in row order; v and target have two entries. Modest finite numeric inputs only. Exact zero suits these integer cases; inspect floating-point discrepancies in context. Predict, use revealing cases, restart and rerun.

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